Sahaj Bhadja

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Summary

Software Engineer pursuing an MSCS at USC with 2+ years of experience building and debugging backend/full-stack systems using Java, Python, SQL, and C++-based logic, with hands-on work in system integration, testing automation, and production troubleshooting.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*
- Built and maintained backend services across **Azure SQL/PostgreSQL** and cloud storage for 500+ users, improving system response from 1.2s to 350ms through schema and query tuning.
- Implemented release validation and reconciliation workflows for 5K+ monthly jobs, improving reliability and traceability of production system changes.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*
- Developed a Flask + React platform handling **10K+ daily requests** with secure ingestion, modular APIs, and robust debugging support across user-facing workflows.
- Built concurrent processing and CI/CD automation with GitHub Actions, reducing turnaround time by 60% while maintaining stable system behavior at peak load.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*
- Built Python-based experimentation and simulation pipelines for autonomous systems, processing **100K+ trajectory logs** and improving long-horizon planning performance by 18%.
- Documented model behavior, validation results, and troubleshooting findings to support reproducible testing and cross-functional integration efforts.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*
- Engineered cross-platform API and ETL workflows using OOP, AWS Glue, and S3; improved response times by 30% and pipeline efficiency by 68% through test-driven fixes.
- Maintained automated validation with Jenkins, PyTest, and Mockito across 3 OS environments, strengthening machine-level troubleshooting and release confidence.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*
- Built a MERN platform with REST APIs and WebSockets supporting 8+ workflows and 2M+ synchronized records with reliable backend integration and system logging.
- Automated deployment/testing with Jenkins and Docker, reducing deployment time by 50% and improving post-deployment support consistency.

Skills

Languages: Java, Python, C++, JavaScript/TypeScript, SQL
Backend & APIs: Flask, FastAPI, Node.js, Express.js, REST APIs, JSON, OOP
Databases: PostgreSQL, MySQL, SQL Server, MongoDB, Azure SQL
Automation & Tooling: Jenkins, GitHub Actions, Docker, Kubernetes, Git, CI/CD, debugging
Systems: Distributed workflows, integration testing, I/O-style validation patterns, technical documentation
Math/ML Foundations: Linear algebra coursework, machine learning pipeline exposure, data validation

Projects

CPU-Scheduling Simulator Live | *C++, React.js, JavaScript*
- Implemented 6 scheduling algorithms with interactive controls and live visualizations across 1K+ simulated process combinations, improving debugging and systems reasoning.

Chat2SQL Code | *Java, TypeScript, SQL, Maven, Docker*
- Built an AI-powered SQL workspace supporting 10+ databases, accelerating query authoring, schema exploration, and structured execution across a Java + TypeScript stack.

CaptionCraft – Image Caption Generator Live | *Transformers, PyTorch, Flask*
- Benchmarked BLIP/GIT/ViT-GPT-2 on a 10K image / 100K caption dataset, improving caption accuracy by 15% (81% to 96%) with reproducible inference workflows.